

Problem 2.9

Show that the function

$$f(t) = 225 - (t - 10)^2$$

cannot be used as an amount function for $t > 10$.

Solution

According to Finan, an amount function (or accumulation function) must satisfy the following three properties:

1. **(P1)** $a(0) = 1$.
2. **(P2)** $a(t)$ is increasing; that is, if $t_1 < t_2$, then $a(t_1) \leq a(t_2)$.
3. **(P3)** If interest accrues for non-integer values of t , then $a(t)$ is a continuous function.

These properties are taken from Finan's book.

To determine whether

$$f(t) = 225 - (t - 10)^2$$

can be an amount function for $t > 10$, it is enough to check property **(P2)**.

Differentiate the function:

$$f'(t) = 0 - 2(t - 10)(1) = -2(t - 10) = 2(10 - t).$$

Now, for $t > 10$, we have

$$10 - t < 0,$$

so

$$f'(t) = 2(10 - t) < 0.$$

Therefore, the function is *decreasing* for all $t > 10$.

This violates property **(P2)**, since an amount function must be increasing (or at least nondecreasing) with time. A decreasing amount function would mean the accumulated value becomes smaller as time passes, which is not allowed for an amount function.

Hence,

$f(t) = 225 - (t - 10)^2$ cannot be used as an amount function for $t > 10$.
